

CrackID: A Benchmark for Crack Re-Identification, and the Appearance Ceiling It Reveals

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Abstract—Automated building-maintenance systems increasingly convert photographs into tracked records: a defect is photographed, a ticket is opened, and follow-up activity accumulates against that record. Any defect re-photographed across visits therefore needs re-identification — a new photo must match the record for that specific physical defect, or open a new one. Wall cracks are a common and difficult case. Yet despite crack detection being mature, we are not aware of a dataset or benchmark for crack *re-identification*, and the natural approach — crop the region and match embeddings — imports an assumption from person and vehicle re-ID that may not transfer: that the object has stable, locally distinctive appearance. We introduce CRACKID: 140 photographs over 17 walls with 212 annotated crack identities, released with an explicit annotation-quality report rather than as unaudited automatic labels. Benchmarking ten matchers across classical keypoints, learned correspondence, general vision embeddings, a re-identification architecture, and a purpose-built crack-shape descriptor, we find pairwise F_1 confined to a 0.054-wide band (0.305–0.359) while Rank-1 spans five times as much and is uncorrelated with it ($\rho = 0.02$). We argue this ceiling is a property of the data, not of any one method. As a geometric reference baseline we register on wall texture with cracks masked out, then Chamfer-match projected crack masks under global assignment: it reaches a detection-and-identification rate of 0.768 at 10% false alarm against 0.414 for the best crop-based method, while answering only 48% of pairs — partial coverage we show by censoring to cost accuracy rather than buy it.

Index Terms—crack re-identification, dataset, benchmark, building maintenance, structural inspection, homography, geometric alignment, annotation quality

I. INTRODUCTION

Automated building-maintenance systems increasingly convert photographic observations into tracked records. A defect is photographed, a ticket or issue is opened, and subsequent activity — status changes, follow-up inspection, resolution — accumulates against that record over the defect’s lifetime. This creates a re-identification requirement for any defect type that gets re-photographed across visits: a new photo must match the existing record for that specific physical defect, or a new record must be opened, without a human reconciling every submission by hand. Wall cracks are a common and difficult case, since the same crack is routinely re-photographed from a different distance, angle, or lighting condition across separate inspection visits.

Crack analysis is a mature computer vision task, but a mature *detection* and *segmentation* task: the public datasets are annotated within single images, and the largest consolidations of them — CrackSeg9k’s 9k images [1], OmniCrack30k’s 30k over twenty sub-datasets [2] — carry no cross-observation

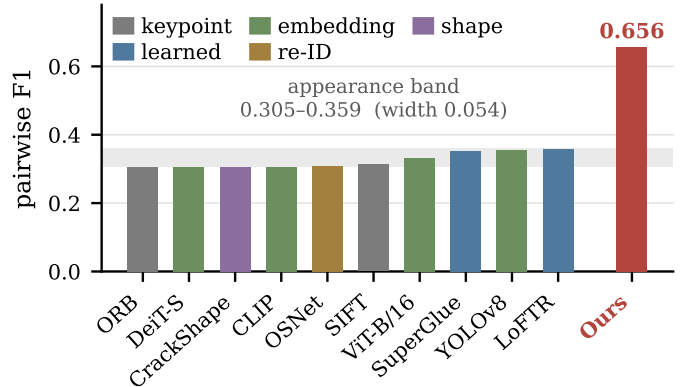


Fig. 1. **The appearance ceiling.** Pairwise F_1 for ten crop-scope matchers spanning five method families, on the CRACKID test split. Every method falls inside a band of width 0.054, irrespective of family, era, or compute budget. The geometric reference baseline (red) leaves the band. Ranking metrics separate these same methods over a 0.30 range; the decision metric does not.

identity labels, because the task they serve does not need any. We are not aware of a dataset or benchmark for crack *re-identification*: deciding which recorded crack a new observation corresponds to, or whether it is new (Sec. II-B).

This gap matters because the natural approach — treat it as instance re-identification, crop the reported region, and match embeddings — carries an assumption from person and vehicle re-ID [3], [4] that does not obviously transfer: that the object being matched has stable, locally distinctive appearance. A crack is thin, low-texture, and locally self-similar; whether crop appearance is actually sufficient to re-identify one has, to our knowledge, never been measured.

We introduce CRACKID to make that measurement possible, and to establish where the difficulty of this task actually lies.

Appearance is not enough, and the flatness says why. On ranking metrics the ten benchmarked matchers separate as expected — Rank-1 spans 0.352 to 0.654. But on pairwise F_1 , the metric measuring the decision a maintenance system actually makes (*same crack or not*, at a threshold), all ten land between 0.305 and 0.359 (Fig. 1). That 0.054-wide band contains ORB and LoFTR, CLIP and OSNet, and a shape descriptor we designed specifically for cracks; the two quantities are statistically uncorrelated ($\rho = 0.02$, $p = 0.96$). A flat band across so diverse a population is evidence about the data, not the encoders.

Building it required solving an annotation problem. Propagating a single click-point identity across a wall’s photographs via homography is the only tractable way to label crack instances at this scale, but it fragments a physical crack into multiple identities whenever propagation breaks, requiring a merge step — and a merge that incorrectly joins two distinct cracks silently converts a hard negative into a free positive and inflates every downstream matching metric at once. We treat this as a first-class part of the contribution rather than a footnote: 45% of CRACKID identities were formed by such a merge, and we report that rate, the gap distribution it bridged, and the agreement between merged and pre-merge labellings, instead of presenting the result as ground truth by construction (Sec. III-C).

Contributions.

- 1) CRACKID, 140 photographs across 17 walls and 212 crack identities, released with an explicit annotation-quality report — merge rate, bridged-gap distribution, chance-corrected agreement against the pre-merge labelling — plus the instrumentation for a human error-rate audit (Sec. III).
- 2) A ten-method benchmark spanning classical keypoints, learned correspondence, general vision embeddings, a dedicated re-identification architecture, and a purpose-built crack-shape descriptor, showing a consistent appearance ceiling that we argue reflects a property of the data rather than of any one method (Sec. V).
- 3) A geometric reference baseline — registration on wall texture with cracks masked out, followed by Chamfer-matched crack masks under global assignment — that substantially exceeds every crop-based method, released alongside the crop-based baselines as part of the benchmark (Sec. VI, VII).
- 4) A protocol for this task’s failure modes: a blur admission gate set by measured impact on registration yield rather than convention, a viewpoint covariate estimated independently of the scoring pipeline to avoid circularity, coverage-aware metrics that keep unanswerable queries, and leave-one-wall-out resampling in place of a point estimate on a small identity count (Sec. IV).

II. RELATED WORK

A. Crack datasets, and where they stop

Automated crack analysis is overwhelmingly a detection and segmentation literature [5], [6], and its datasets follow: CrackTree, CFD, CRACK500, DeepCrack, GAPs and their successors annotate cracks *within* single images, and the large consolidations merging them [1], [2] inherit that scope. None carries an identity label linking a crack in one photograph to the same physical crack in another, because segmentation does not need one. They answer *where is there a crack in this image*, never *is this the crack from last time*, and the distinction is not cosmetic: a detector perfect on every visit still yields an incoherent defect history if the visits cannot be linked. Multi-view pipelines aggregate across images of one

inspection [7] but establish no instance correspondence, so they cannot support temporal tracking either.

B. Multi-temporal crack monitoring: the nearest neighbour

Structural health monitoring does compare crack images across time, with machinery close to ours. Parente *et al.* [8] track crack width from repeated captures; Harb *et al.* [9] use temporal sequences to improve crack *segmentation*; Sun and Chang [10] perspective-correct handheld time-lapse images so evolution can be measured under field distortion. RANSAC-estimated homography is the standard tool throughout, and it appears in patents too, matching a new crack image against registered ones via SIFT [11].

The task is different in a way that matters. Throughout this work the *identity of the crack is given by construction* — a fixed camera, a marked location, or an operator guided back to a previous viewpoint supplies the correspondence — and the open question is the crack’s width or extent. Accordingly the metrics are measurement metrics (width, area, spine-length error) rather than association metrics; the subject is one known crack rather than a wall of candidates; and no dataset is released on which one association method could be compared against another. The patent case is telling: its SIFT-and-homography pipeline exists to guide an operator back to the same camera pose, which *avoids* the re-identification problem rather than solving it.

What is missing is the decision an automated maintenance record requires: given a new photograph of a wall carrying several cracks, which recorded identity does each observation belong to, and which are new? That is an open-set association problem over a gallery, it has no benchmark, and it is what CRACKID supplies.

C. Object re-identification

Re-identification matches an instance across captures [4]. Handcrafted local descriptors [12]–[14] are efficient and interpretable but need repeatable, distinctive keypoints — precisely what hairline cracks on flat render fail to provide. Learned correspondence replaces descriptor matching with attention: SuperGlue [15] refines detected keypoints through a graph network and an optimal matching layer, and LoFTR [16] is detector-free and dense. Both are markedly more robust, and both are pairwise, so cost grows with gallery size. Embedding methods map a region to a vector [17]: ViT and DeiT [18], [19], CLIP [20], detector backbones [21], and OSNet [3], which is designed for instance-level discrimination.

Our benchmark evaluates all of these. Our finding is not that any one of them is weak, but that their differences collapse on the decision metric — a result that only becomes visible when the population is broad enough and the metric is the deployable one rather than a ranking statistic.

D. Damage association in inspection pipelines

The closest integrated systems come from vehicle-damage assessment. Maiano *et al.* [22] combine filtering, detection, and OSNet-based damage re-identification for insurance

claims. Commercial vehicle inspection platforms rely on controlled capture rigs. In both cases association is posed over cropped damage regions and evaluated where acquisition is curated. Our results suggest the crop scope is itself the limiting choice, and that a geometry-first formulation — available whenever the damaged object is a rigid surface with texture of its own — changes what is achievable.

E. Geometric alignment

Planar registration is classical [23], with RANSAC-family robust estimation [24] and well-understood sensitivity to control-point configuration [25]; direct photometric alignment (ECC) is the standard fallback where keypoints fail. Our contribution is not a new estimator but the observation that in crack correspondence the alignment target and the matching target should be *disjoint*: the homography must be driven by the wall precisely because the crack is what we are trying to identify. Masking cracks out of detection is what makes the projected position an independent identity cue rather than a restatement of appearance similarity.

III. CRACKID

A. Capture and composition

We photographed 17 exterior walls at one site over four half-days with two consumer handsets, yielding 140 photographs at 4080×3072 . The walls span two surface families that behave very differently under keypoint matching: stippled render, which is densely textured, and smooth painted plaster photographed at cornice junctions, where the crack is frequently the only structure in frame. Walls are split disjointly into 6 validation walls (66 photographs, 66 identities) and 11 test walls (74 photographs, 146 identities); no wall appears in both.

Statistical power is governed not by the identity count but by identities visible in *two or more* photographs, since only those form a query with an answer: 18 in validation, 19 in test, 37 in total. This is the single largest constraint on what the benchmark can conclude, so we state it here rather than in a footnote (Sec. VIII).

B. What the dataset measures, and what it does not

Each wall was captured in a single continuous walk-around. Consecutive frames are a median of 3 s apart (maximum 20 s), and a wall’s entire capture spans a median of 29 s (maximum 81 s). Every query and its gallery therefore share one illumination condition, one weather state, and one sensor configuration.

We state the consequence plainly. CRACKID in its present form measures **viewpoint and scale correspondence**: given a crack seen from one camera position, find it from a different position, distance, and angle. It does **not** yet measure correspondence across separate visits, which is the deployment case the maintenance-record setting implies. We therefore use *correspondence* rather than *re-identification* for the measured property, and treat every figure here as an upper bound on multi-visit performance. Sec. VII-E quantifies the residual viewpoint variation the data does span, using an estimator the scoring pipeline never sees.

C. Annotation, and why it needs auditing

Labelling crack instances by hand across 140 high-resolution photographs is not tractable, so identities are seeded automatically. An annotator places one click point per distinct physical crack; the point is propagated between consecutive photographs through an ORB homography and a one-to-one assignment; and identities whose components appear to be fragments of one physical crack are then merged.

The merge step is where the risk concentrates. A merge that wrongly joins two distinct cracks converts a hard negative into a free positive and inflates mAP for *every* method at once — so it cannot be caught by comparing methods against each other. We therefore report its extent rather than assume its correctness.

95 of the 212 identities (45%) were formed by a merge — 70 of 146 on test (48%), 25 of 66 on validation (38%) — bridging a median maximum gap of 228 px on test and 286 px on validation (p90 308 and 317 px, max 319 and 320 px). The merge restructured the partition almost completely: the adjusted Rand index between the final labelling and the pre-merge automatic output — a measure of how much the merge changed the partition, not of inter-annotator agreement — is 0.080 over 1,051 matched points. That is the same fact as “45% absorbed more than one seed,” seen from the other side. It is not an argument against merging, which is what makes the identities physically meaningful; it is why an audited error rate must exist before mAP on these labels is treated as exact.

We release the instrumentation for that audit rather than the audit itself: per identity merge provenance, a reproducible stratified sample over merged and unmerged identities with rendered contact sheets, and scoring that turns human verdicts into an error rate with a Wilson interval. An automatic triage flags the 57 identities with the widest bridged gaps, where disagreement will concentrate. **The verdicts are not yet filled in**, so we report no sampled error rate and no inter-annotator statistic and claim none; Sec. VIII states what follows.

IV. BENCHMARK PROTOCOL

A query is one annotated crack instance in one photograph. The gallery is restricted to **the query’s own wall**: the hard negatives are other cracks in the same material under the same light, which is the honest setting and markedly harder than a cross-wall gallery. A query’s own photograph and its own capture session are excluded. This yields $n_Q = 280$ scored query instances on test.

Three protocol choices are deliberately unfavourable to the methods being measured, including ours. **Unanswerable queries are retained** — a query whose identity has no gallery counterpart is not deleted, since that is exactly the open-set case a maintenance system faces whenever a crack has been repaired or is genuinely new; such queries can contribute false alarms and never successes. **Thresholds are calibrated on validation**: we report an oracle-thresholded F_1^* (best threshold on test, which flatters every method) and $F_1^{\text{@}v}$, thresholded on the held-out validation walls; where only one column is available for a baseline it is the oracle variant, so the

comparison is biased *towards* the baselines. **Missing scores are charged, not skipped:** pairs a method cannot register take the failure score and rank last, with the query still counted in every metric (Sec. VII-C).

A. Metrics

We report Rank-1, Rank-5 and mAP for ranking, resolving ties in expectation over random tie-breaks rather than by array order, which otherwise rewards methods emitting many equal scores. The headline metric is the open-set **detection-and-identification rate at 10% false alarm** (DIR@FAR=0.1): a known query is correct only if its top-ranked gallery entry is above threshold *and* correct, while an unknown query is a false alarm whenever anything clears threshold. We also report **pairwise F_1** over all valid pairs independent of ranking, and **assignment F_1** , a one-to-one matching per image pair — the deployed operation, since two cracks in today’s photograph must not both claim the same record.

Confidence is reported by resampling **walls**, not queries: queries from one wall share its material, lighting, and camera pass and succeed or fail together, so a query-level interval is badly over-optimistic. With 11 test walls that interval is wide, and we regard its width as the honest statement of what 19 multi-photo identities support.

V. THE APPEARANCE CEILING

A. Setup

We evaluate ten crop-scope matchers under the identical protocol of Sec. IV, on identical inputs, scored by identical code from saved score matrices. The population is deliberately broad, spanning five families:

- **Handcrafted keypoints:** SIFT [12], ORB [14]. Similarity is the inlier count after ratio test and RANSAC verification.
- **Learned correspondence:** SuperGlue [15], LoFTR [16]. Similarity is the confidence-weighted count of predicted correspondences.
- **General visual embeddings:** ViT-B/16 [18], DeiT-S [19], CLIP ViT-B/32 [20], and a YOLOv8 detector backbone. Similarity is cosine distance between L2-normalised features.
- **A re-identification architecture:** OSNet [3], trained for instance-level discrimination.
- **A purpose-built crack-shape descriptor** (CRACKSHAPE), the strongest crop-scope baseline we could construct *knowing the domain*: it PCA-aligns crack pixels to remove in-plane rotation, resamples the centreline into 48 bins, normalises perpendicular offsets by crack length to remove scale, and takes the magnitude spectrum of the offset profile — invariant to traversal direction — with scale-free tortuosity, elongation, relative thickness, and turning-angle spread appended.

CRACKSHAPE matters to the argument: if the ceiling were an artefact of generic encoders meeting a specialised structure, a descriptor designed around crack geometry should escape it. It does not.

B. Result

Table I reports all ten methods. The pattern is the point. **Pairwise F_1 is flat.** Every method lands in $[0.305, 0.359]$: mean 0.324, standard deviation 0.023, coefficient of variation 0.068. The span is 0.054, or 16.7% of the mean. By comparison Rank-1 spans 0.302 (54.6% of its mean, CV = 0.164) and DIR@FAR=0.1 spans 0.396 (CV = 0.655) over the same ten methods on the same data. Whatever separates these methods, it does not separate their ability to decide whether two crops show the same crack.

Ranking ability does not transfer to decision ability. Across the ten methods, Rank-1 and pairwise F_1 are essentially uncorrelated (Spearman $\rho = 0.018$, $p = 0.96$). OSNet ties for the best Rank-1 (0.654) and sits near the *bottom* of the pairwise band (0.309); SuperGlue ranks sixth on Rank-1 (0.555) yet achieves the best open-set rate of any baseline (0.414 versus OSNet’s 0.168). A method that orders a gallery well is not thereby a method that can say *yes, that is the same crack* — and it is the second capability an inspection system needs.

Compute buys nothing. Cost per query spans $2590\times$, from 0.003 s (CRACKSHAPE) to 8.33 s (LoFTR). Its rank correlation with Rank-1 is $\rho = 0.018$ ($p = 0.96$) and with pairwise F_1 is $\rho = 0.497$ ($p = 0.14$, not significant at $n = 10$). LoFTR costs $1300\times$ OSNet per query and ranks below it on Rank-1, Rank-5, and mAP.

The absolute level is the real finding. A pairwise F_1 near 0.33 is not a merely-imperfect system; it is unusable for maintenance bookkeeping, where a wrong association silently corrupts a defect history. The ceiling is low as well as flat.

C. Why we read this as a property of the data

Three observations support a data-level rather than encoder-level explanation. The band contains methods with no shared inductive bias — sparse keypoints, dense attention, ImageNet features, language-image pretraining, a re-ID metric objective, hand-designed shape statistics — and a common limitation of such a population sits more parsimoniously in the signal than in five independent architectural failures. The domain-specific descriptor lands at the bottom of the band. And the failure mode is structural: two distinct hairline cracks under one illumination are routinely more similar than two views of one crack at different scales.

We state the limits of this inference. Ten methods are a sample, not a proof; $n=10$ makes the correlations descriptive rather than inferential; and a model trained on a large corpus of crack instances — which does not exist — might yet break the band. What the result establishes is that *the obvious moves have been tried and land in the same place*, reason enough to change the formulation rather than the encoder.

D. A designed viewpoint ablation

The evidence above is observational: the viewpoint variation is whatever the capture produced. To test sensitivity by design, we warp each admitted photograph, its mask, and its label points by a homography with *exactly known* scale, rotation, and tilt, and query the wall’s *unmodified* real gallery

TABLE I

MAIN RESULT, TEST SPLIT, $n_Q = 280$ QUERY INSTANCES OVER 11 WALLS. ALL METHODS SEE IDENTICAL INPUTS AND ARE SCORED BY IDENTICAL CODE. F_1^* IS ORACLE-THRESHOLDED ON TEST; FOR THE PROPOSED METHOD WE ADDITIONALLY REPORT $F_1^{\text{@}v}$, THRESHOLDED ON THE HELD-OUT VALIDATION WALLS, WHICH IS THE DEPLOYABLE NUMBER. *scored* IS THE FRACTION OF VALID QUERY-GALLERY PAIRS RECEIVING A FINITE SCORE; UNSCORED PAIRS RANK LAST AND THE QUERY IS STILL COUNTED (SEC. VII-C). COST IS WALL-CLOCK SECONDS PER QUERY ON ONE WORKSTATION. BEST CROP-SCOPE VALUE PER COLUMN IN **BOLD**.

Method	Family	R@1	R@5	mAP	DIR@FAR=0.1	F_1^*	assign- F_1	scored	s/query
<i>Crop scope</i>									
ORB [14]	keypoint	0.352	0.759	0.364	0.018	0.305	0.325	1.00	0.056
SIFT [12]	keypoint	0.432	0.829	0.417	0.036	0.313	0.324	1.00	0.119
SuperGlue [15]	learned	0.555	0.893	0.453	0.414	0.353	0.352	1.00	5.693
LoFTR [16]	learned	0.549	0.824	0.441	0.336	0.359	0.401	1.00	8.325
YOLOv8 backbone	embedding	0.561	0.821	0.453	0.157	0.356	0.364	1.00	0.016
ViT-B/16 [18]	embedding	0.654	0.893	0.511	0.286	0.331	0.392	1.00	0.851
DeiT-S [19]	embedding	0.621	0.879	0.497	0.250	0.305	0.381	1.00	0.259
CLIP ViT-B/32 [20]	embedding	0.604	0.850	0.471	0.143	0.307	0.367	1.00	0.228
OSNet [3]	re-ID	0.654	0.907	0.520	0.168	0.309	0.388	1.00	0.006
CRACKSHAPE (ours)	shape	0.554	0.825	0.442	0.082	0.305	0.346	1.00	0.003
<i>Full-image scope</i>									
REG+CHAMFER (ours)	geometry	0.952	0.992	0.886	0.768	0.656/0.582^{@v}	0.784	0.48	27.92

with it. Only the geometry between the views is synthetic; crack appearance, texture, and sensor noise are real, and the construction is self-checking — decomposing the transform recovers the requested parameters to floating point.

There is almost no response. From the identity transform (SIFT Rank-1 0.469, ORB 0.456, $n=207$), rotating every query by 45° costs SIFT 0.051 and gains ORB 0.046 ($n=201$); a 60° tilt costs SIFT 0.037 and gains ORB 0.033 ($n=192$). A $3\times$ scale change moves SIFT 0.469 $\rightarrow$ 0.424, but is confounded: upscaling pushes labelled points out of frame, so n falls $207 \rightarrow 66$ and the survivors are a more central subset.

We read this as corroboration, not a separate finding. A merely *viewpoint-limited* method should degrade as the transform grows; these do not, because at Rank-1 ≈ 0.46 they already sit near a floor set by crack self-similarity. Two limits: only the two cheapest baselines were run, and the transforms are exactly affine by construction, so this says nothing about projective distortion.

VI. A GEOMETRIC REFERENCE BASELINE

The ceiling motivates a change of scope rather than a better encoder. A crack is a mark at a fixed location on a rigid, near-planar surface, and that surface carries texture the crack does not. We therefore identify a crack by *where it is on its wall* after the two photographs have been aligned.

We release this as a *reference baseline* for CRACKID, not a finished system: deliberately simple, with no learned component beyond the crack segmenter every method shares, marking where the geometric formulation sits so later work has something to beat. It has four stages.

A. Alignment driven by non-crack texture

Given photographs I_a, I_b of one wall with crack masks M_a, M_b , we estimate the homography H with $x' \sim Hx$

mapping I_a into I_b 's frame. Walls are planar, so a homography is the correct model rather than an approximation.

The design decision that matters is **what drives the fit**. We dilate the crack masks by 15 px and *exclude* those pixels from detection, so H comes from wall texture alone. This is not an optimisation but what makes the resulting position an independent identity cue: were cracks to participate in the alignment, a long crack could dominate the correspondence set and pull the fit toward superimposing the wrong crack, and the projected position would restate appearance similarity rather than provide evidence against it.

Detection runs on copies downscaled to 1600 px under CLAHE equalisation, so exposure differences between captures do not suppress matches; H is rescaled to full resolution afterwards. We use SIFT (up to 8000 features), a 0.75 ratio test, and MAGSAC at 4 px requiring 25 inliers.

B. A cascade for two surface families

One front end does not serve both surfaces in this data, and the failure is categorical rather than marginal: on the smooth-plaster walls, crack-masked SIFT succeeds on *exactly zero* pairs. We therefore try three front ends in order.

- 1) **SIFT on masked wall texture**. Correct for stippled render.
- 2) **SIFT with cracks restored**, triggered only when masking starves detection below 300 keypoints in either image. On smooth plaster at a cornice junction the crack is the only feature in frame, and masking it leaves nothing. This stage accepts the circularity that stage 1 avoids, and we record which front end produced each result so the trade is auditable.
- 3) **ECC**, direct photometric alignment needing no keypoints at all, over a $320 \rightarrow 640 \rightarrow 1280$ px pyramid opening with a translation model. It aligns on large-scale shading, the only signal surviving on flat paint.

Every candidate H passes a sanity check rejecting the degenerate matrices RANSAC otherwise returns: we warp the image corners and reject if the quadrilateral is mirrored or folded, if its area scale leaves $[1/16, 16]$, or if a side collapses. The fit’s inlier RMS is retained to set downstream tolerances rather than fixing a constant.

C. Projection and scale-free Chamfer agreement

Crack instances are extracted as connected components. Instance pixels from I_a are projected into I_b by H and compared to instances of I_b .

Chamfer distance, not IoU, is the primary criterion. Cracks are a few pixels wide, so a 3px registration error drives IoU to nearly zero for a pair that is in fact correctly aligned. We use the symmetric mean distance between pixel sets, computed over the union bounding box of the two instances with a cheap rejection when boxes are far apart.

Raw Chamfer is a distance in *pixels*, and every photograph pair carries its own independently solved homography, so its scale differs per pair and per crack size: 3px of error on a 40px crack and on a 400px crack are not the same evidence. Since DIR@FAR, pairwise F_1 , and the assignment all apply *one global threshold*, a raw-pixel score would make them measure calibration rather than accuracy. We therefore score with **Chamfer coverage**, the symmetric fraction of each instance’s pixels lying within τ of the other,

$$C(a, b) = \frac{1}{2} \left[\frac{|\{p \in a : d(p, b) \leq \tau\}|}{|a|} + \frac{|\{q \in b : d(q, a) \leq \tau\}|}{|b|} \right] \quad (1)$$

which is bounded in $[0, 1]$ and scale-free, so a threshold learned on validation transfers to test. We set τ from the registration’s own inlier RMS for that pair where available.

D. Global assignment and explicit non-answers

Correspondence is solved as a one-to-one assignment per image pair by the Hungarian algorithm with an explicit rejection threshold, not by per-query argmax — which would permit two cracks in I_a to both claim the same crack in I_b , not a physically admissible reading of a surface. Unmatched instances are reported as *disappeared* or *newly appeared* rather than forced to a nearest neighbour.

This is where the geometric formulation earns its place. Because the two photographs are aligned, the system can project a recorded crack into the new image and observe that the region is visible and the crack is not in it. That is a **licensed negative** — positive evidence of absence — and it is categorically stronger than the low similarity score that is the only negative a crop embedding can produce. It is also the mechanism behind the open-set gap reported next.

VII. RESULTS

A. Main comparison

Table I places the geometric reference baseline against all ten crop-scope methods. Relocating identity into wall geometry moves every metric decisively: Rank-1 rises from 0.654 to 0.952, mAP from 0.520 to 0.886, assignment F_1 from 0.401

to 0.784. Pairwise F_1 leaves the appearance band entirely, at 0.656 against a ceiling of 0.359 — and the *validation-calibrated* $F_1^{\text{cv}} = 0.582$ still exceeds every baseline’s *oracle* value by $1.62\times$, a comparison biased in the baselines’ favour.

B. The open-set gap is the substantive result

The clearest separation is on DIR@FAR=0.1: 0.768 against 0.414 for the best baseline (SuperGlue), a $1.86\times$ improvement, and $4.6\times$ over OSNet, the strongest baseline by Rank-1.

This is the licensed negative, quantified. The metric charges a false alarm whenever an unanswerable query produces an above-threshold match, and crop methods cannot decline: they return a similarity to their nearest neighbour, which on a wall of self-similar hairline cracks is often high. Our method instead aligns the photographs, projects the query into the gallery frame, and finds the location visible and empty — holding a high detection rate at a false-alarm budget where appearance methods have spent theirs. Compute does not help: SuperGlue’s 0.414 costs 5.7 s per query, OSNet’s 0.168 costs 0.006 s.

C. Partial coverage is a penalty, not a subsidy

The baseline returns a finite score for 48% of valid pairs; every crop method returns one for all of them. The intuitive reading — that answering only easy pairs flatters the metrics — is wrong here, and we tested it rather than asserted it. Unscorable pairs take the failure score and rank *last*, with the query still counted. Censoring a complete score matrix artificially, Rank-1 fell $0.422 \rightarrow 0.27$ at 34% coverage, and censoring whole image-pair blocks — registration’s actual failure structure — was no more forgiving than censoring cells uniformly.

The unscored fraction is not one phenomenon: part is pairs that could not be registered, a genuine non-answer; the rest is pairs that registered and in which the query crack projects *outside* the gallery photograph — a confident “not in that image” no crop method can produce. The artefacts report the split.

D. Failures are structured by wall, not scattered

Registration succeeds on 97 of 301 same-wall test pairs (32.2%) without an admission gate, but the average conceals the mechanism (Fig. 2): walls 07, 08 and 12 register at 100%, wall 11 at 83%, wall 09 at 52%, while walls 16, 14 and 13 sit at 3%, 6% and 11%, and walls 10 and 17 at zero.

This bimodality is why we resample walls rather than queries, and why a single coverage number misleads. Leave-one-wall-out on the cheapest crop descriptor makes the variance concrete: CRACKSHAPE scores Rank-1 0.495 ± 0.329 and mAP 0.464 ± 0.294 over 14 folds — a wall-to-wall standard deviation of 0.33 on a mean of 0.50, which is why we claim no fine-grained separation among baselines.

E. What predicts a registration failure

Because the failures are structured, we model them. We fit one joint logistic regression over all 301 test pairs so

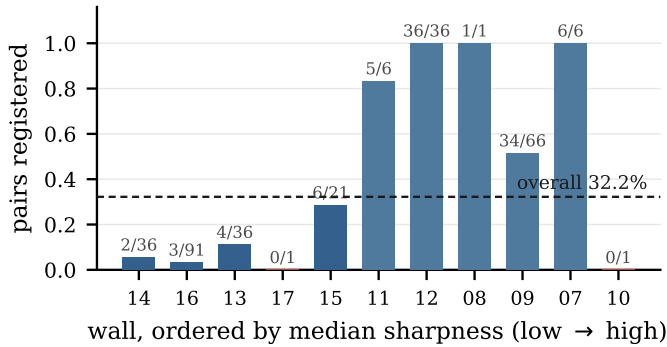


Fig. 2. Registration coverage per test wall, ordered by median sharpness. Failure is categorical rather than gradual and largely tracks image quality — the walls at left are the blurred smooth-plaster captures. Walls 10 and 17 contribute one pair each; wall 10 is the counterexample, sharp yet unregistered.

each effect is controlled for the other. Sharpness dominates: $\log(1 + \text{sharpness})$ has Wald $z = 8.23$ ($p = 1.8 \times 10^{-16}$), with median sharpness 133.8 among registered pairs against 7.2 among failures. Frame gap remains significant once sharpness is controlled ($z = -5.03$, $p = 4.8 \times 10^{-7}$) but is far smaller. Registration failure in this benchmark is primarily an image-quality event, not a viewpoint event.

Measuring the viewpoint the data spans requires care: a covariate derived from the method’s own homography exists only where that method succeeded, so stratifying on it would be circular. We therefore estimate viewpoint with an *independent* front end the scorer never uses — AKAZE/ORB on unmasked images, falling back to dense ECC — decomposed into isotropic scale change, in-plane rotation, and the out-of-plane tilt implied by foreshortening. It solves 139 of 301 pairs (46%) against the scorer’s 97 (32%), and 51 of those are pairs the scorer *failed* on: a quarter of all registration failures now carry a viewpoint measurement the scorer did not produce. On the 32 pairs both estimators solve, median $|\log\text{-scale}|$ disagreement is 0.106 (11.2%) and median rotation disagreement 2.91° , justifying pooling.

The resulting spread (Fig. 3) is substantial: median scale change $1.42\times$ (p90 $2.26\times$, max $5.20\times$), median absolute in-plane rotation 6.9° (p90 39.6°), and median out-of-plane tilt 25.6° (p90 75.3°). So although captures are seconds apart, they are not trivially similar in geometry.

We also use this covariate to rule out a tempting shortcut. Frame-index gap is *not* a viewpoint proxy: over the registered pairs it correlates with in-plane rotation ($\rho = 0.31$, $p = 2 \times 10^{-4}$) and tilt ($\rho = 0.35$, $p = 2 \times 10^{-5}$) but not with scale change ($\rho = 0.10$, $p = 0.24$). A frame-gap sweep is therefore largely a camera-roll sweep, and roll is the one component a homography absorbs exactly. We report frame gap as a near-duplicate control and build no invariance claim on it.

F. Ablation: the third front end

The ECC stage exists for the smooth-plaster walls on which keypoint matching finds nothing. Disabling it and leaving the pipeline otherwise identical gives Rank-1 0.920, Rank-5

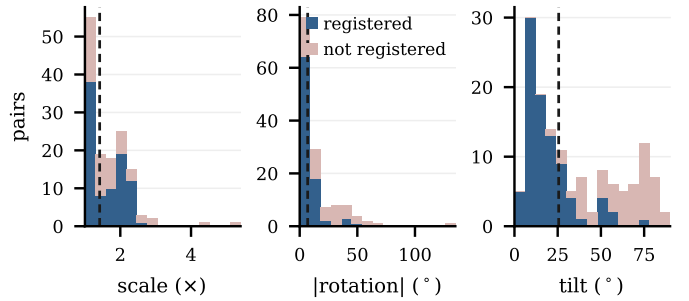


Fig. 3. Viewpoint spanned by the benchmark, estimated by a front end the scoring pipeline never uses, so that stratification is not circular. Dashed line marks the median. Registration failures (pale) are not concentrated at extreme viewpoints — consistent with the logistic fit, in which sharpness rather than viewpoint dominates.

0.956, mAP 0.862, DIR@FAR=0.1 of 0.768, and coverage 0.50, against 0.952, 0.992, 0.886, 0.768, and 0.48 for the full cascade.

This ablation has an unusual provenance we state plainly: the disabled condition was not constructed but discovered. The ECC stage had been silently inert — its pyramid began where convergence always failed, and a failed level aborted the routine — so earlier measurements were taken with a dead third stage. Repairing it makes the disabled condition an exact ablation rather than an approximation.

The result is instructive in a way we did not expect. Enabling ECC raises every ranking metric and assignment F_1 ($0.769 \rightarrow 0.784$) while coverage moves $0.50 \rightarrow 0.48$ — essentially flat, and slightly *down*. The stage is therefore not buying accuracy by admitting marginal alignments; it fixes pairs the keypoint stages resolved badly and correctly declines others. The open-set rate is unchanged, locating ECC’s benefit in ranking rather than in the accept/reject decision.

G. Admission policy: making image quality explicit

Since sharpness dominates registration failure, whether a capture is *admitted* is a design decision, not a preprocessing detail: hiding it would let any method be tuned by silent input selection. We apply one gate identically to every method, before extraction. No gate keeps every wall pairable, so the question is which loses least. At variance-of-Laplacian ≥ 10 (quarter-scale grey), 21 of 74 test captures are rejected, one wall (17, already the least sharp) drops to a single photograph, and the registration rate on what remains nearly doubles, $32.2\% \rightarrow 56.1\%$. A stricter gate at 25 reaches 68.3% but costs three more walls any same-wall pair at all. We adopt 10, the loosest threshold costing at most one wall. Table I reports the *ungated* protocol, comparing all methods on all captures.

H. Cost

At 27.9s per query the baseline is $4300\times$ slower than OSNet, with two qualifications. The cost is dominated by registration, which is per *image pair*: one homography serves every crack in it, so on a wall with many recorded cracks the amortised cost falls proportionally. And it is within an order of

magnitude of LoFTR (8.3s), a crop-scope method with none of the open-set benefit.

VIII. THREATS TO VALIDITY

We state the limits in the body rather than a closing paragraph, because two of them bound the central claim.

A. Scope is uncontrolled between the two formulations

The proposed method reads full photographs; the baselines read crops. That follows from the design — homography estimation needs wall-level context a crop cannot supply — but it means Table I varies *two* things at once: identity cue (appearance versus geometry) and input scope (crop versus full image). The result establishes that a geometry-first system on full images substantially outperforms crop-scope appearance matching here; it does not by itself establish that geometry rather than context is responsible.

Two controls separate these and ship with the benchmark: a *crop-plus-context* condition widening the crop until baselines see comparable surrounding wall, and a *crack-erased* condition that inpaints the crack away, so any remaining performance is attributable to context alone. Both are designed but unrun — they require regenerating score matrices for all ten baselines — and we prefer to leave the gap visible. The ceiling result is unaffected, comparing crop methods only against each other.

B. The effective sample is 19 identities

The test split’s 280 query instances come from 19 identities visible in two or more photographs, over 11 walls. Query count is not sample size: queries from one wall share material, illumination, and one camera pass. Hence we resample walls and report the leave-one-wall-out spread (± 0.33 Rank-1 for CRACKSHAPE).

This benchmark can therefore support a large structural claim — a 0.35 gap in DIR@FAR, a formulation that leaves a band ten methods sit inside — but not fine-grained ranking among baselines. Differences of a few points between two crop methods in Table I should be read as uninformative.

C. The annotation audit is instrumented but not completed

The human verdicts that would turn the provenance of Sec. III-C into a sampled error rate with a confidence interval have not been collected, and no second annotator has labelled a subset. Until both exist, every mAP figure here — ours and the baselines’ alike — is conditional on the merge policy being sound. Relative comparison is more robust than absolute level, since the merge affects all methods identically, and the appearance ceiling is a comparison among methods rather than an absolute claim.

D. Segmentation quality is not reported

Crack masks come from a learned segmenter and are treated as given. Mask errors propagate into instance extraction and hence Chamfer agreement, and we do not quantify that path: hand-drawn reference masks do not exist for this data. The released code contains a sensitivity sweep over minimum-area

and closing parameters, checkable once reference masks are drawn.

E. Site, surface, and device scope

All captures come from a single site over four half-days, two handsets, and two surface families. We claim a benchmark and a baseline for these conditions, not a general solution. The baseline assumes a near-planar surface with texture that is not the crack, and should degrade on curved or featureless surfaces, and where the surrounding texture itself changes between visits.

F. The measurement that would change the most

One thing would convert the central claim from inferred to designed: a second capture session, weeks later, at marked distances and angles. It would make the protocol cross-visit rather than viewpoint-only, supply exact rather than estimated viewpoint ground truth, and test the condition under which we expect geometry to widen its advantage — appearance drift with position preserved.

IX. CONCLUSION

Maintenance systems that keep a record per physical defect must re-identify that defect when it is photographed again. For wall cracks there was no dataset on which to test whether that is possible, so we built one: CRACKID, 140 photographs over 17 walls with 212 crack identities, released with its annotation provenance stated rather than assumed — 45% of identities were formed by an automatic merge, and we report that rate rather than presenting the labels as ground truth by construction.

The benchmark’s main finding is negative and, we think, structural. Ten matchers across five families all decide *same crack or not* at a pairwise F_1 between 0.305 and 0.359 — a 0.054-wide band across two decades of method development and a $2590\times$ range in compute — while Rank-1 spans five times as much and is uncorrelated with it. Ordering a gallery well and deciding an identity are not the same capability, and only the second maintains a record.

Our geometric reference baseline shows the ceiling is not a property of the task. A homography from non-crack wall texture, crack masks projected into a common frame, and scale-free Chamfer agreement under global assignment reach DIR@FAR=0.1 of 0.768 against 0.414, while answering only 48% of pairs. The advantage has a specific source: an aligned pair supports a *licensed negative* — project the recorded crack, look, observe it is not there — where a crop embedding can only report low similarity. We expect that to matter for any defect marking a rigid, textured surface. Two bounds remain, and both are addressable: a second capture session weeks later turns CRACKID from a viewpoint benchmark into a cross-visit one, and the annotation audit is instrumented and awaits its verdicts.

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